Plate Reader Data Pipeline Application for TERM Lab

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Background

Client

Tissue Engineering and Regenerative Medicine (TERM) Laboratory

Focus: Cell culture-based assays using plate readers

Pain points:

- Manual workflows
- Time-consuming
- Inconsistent analysis
- Poor documentation/not reproducible
- Difficult for new lab members
- Reliance on proprietary software (GraphPad Prism)

TERM — Tissue Engineering & Regenerative Medicine Laboratory

TERM Lab Workflow

1. Inport raw CSV into Prism
2. Manually enter plate layout
3. Normalize data by hand
4. Basic statistical tests (often inaccurate)
5. Export student-generated figures
6. Write methods separately

Objective

Build a fully reproducible, GUI-based R pipeline that:

- Automates normalization and statistics
- Generates publication-ready figures
- Produces a reproducible PDF report
- Serves as a tool for both research and teaching

Plate Reader

Physical plate: typically 96 wells (A–H rows, 1–12 columns)

- Each well contains a sample, blank, or control
- The plate reader: scans wells individually and measures signal
  - e.g light absorbance, fluorescence, luminescence
- Output: table of raw signal values corresponding to each well

Raw .csv output:

- Exported directly from plate reader software (CSV/Excel)
- Structured as an 8×12 grid
- Each cell = one numeric measurement
- No labels included

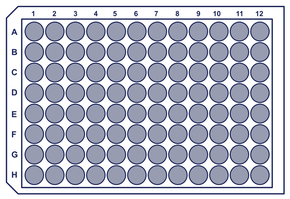


Figure 1: Microplate diagram

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B		11755	13591	11440	12549	12117	11778	1337	1344	1360		
C		9496	8828	9766	9429	9152	8845	1396	1377	1378		
D		5746	5885	5867	5364	6834	7301	1378	1377	1383		
E		4587	5936	5255	6319	6801	6345	1381	1407	1389		
F		3889	3717	4019	4164	4224	4072	1382	1394	1405		
G		7694	7071	7410	7434	5187	5398	1349	1365	1343		
H												

Figure 2: Sample plate data

Sample outputs

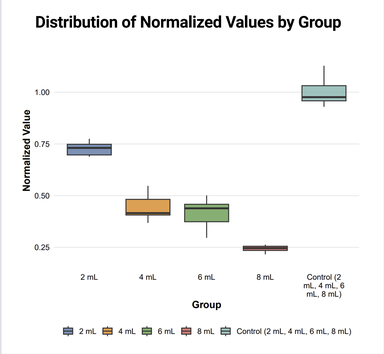


Figure 3: Example boxplot graph output by the application

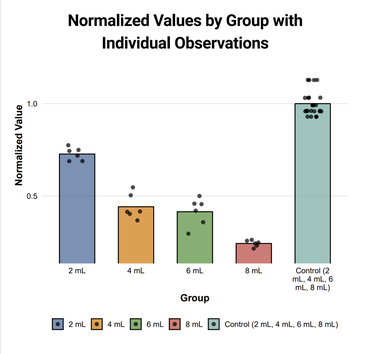


Figure 4: Example bar chart output by the application

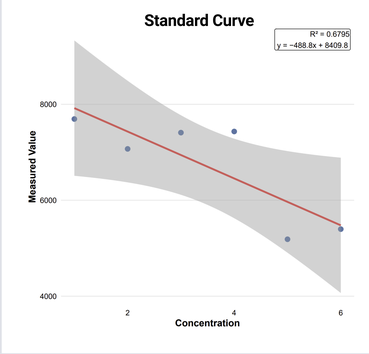


Figure 5: Example standard curve output by the application

2 mL	4 mL	6 mL	8 mL	Group
5.889666e−10	NA	NA	NA	4 mL
5.913778e−11	4.503579e−01	NA	NA	6 mL
7.813921e−17	1.695535e−06	1.859824e−05	NA	8 mL
3.921102e−12	8.497032e−23	2.011614e−23	1.354203e−27	Control (2 mL, 4 mL, 6 mL, 8 mL)

Figure 6: Sample pairwise t-test outputs from application analysis

Term	Degrees_of_Freedom	Sum_of_Squares	Mean_Square	F_Statistic	P_Value	Significance
plot_group	4	4.2584237	1.064605935	282.6396	2.896845e−30	***
Residuals	43	0.1619662	0.003766655	NA	NA	ns

Figure 7: Sample ANOVA test output from application analysis

Extra Features

User Guide: Built-in step-by-step guide explaining workflow, analysis choices, and example use cases

Bug Reporting Tool: Simple in-app reporting system to log issues and improve the development feedback loop

Light/Dark Mode Toggle: Header-based theme switch for improved accessibility and user preference

Acknowledgements

The development of this project was made possible through the guidance, collaboration, and support of:

- Dr. Robert Buscaglia
- Christopher Groberg
- TERM Lab

References: Chang W, Cheng J, Allaire J, Sievert C, Schloerke B, Aden-Buie G, Xie Y, Allen J, McPherson J, Dipert A, Borges B (2025). _shiny: Web Application Framework for R_. R package version 1.12.1, <https://CRAN.R-project.org/package=shiny>.

GitHub Repository

Application Workflow

Screen 1: Upload

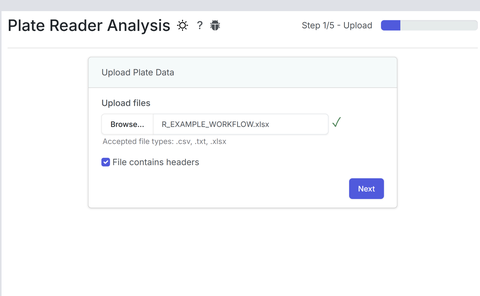
Purpose: Import and initialize experimental data

Input: Raw plate reader file

Features:

- Upload plate data (CSV/Excel/TXT)
- Support multi-plate uploads
- Input validation ensures correct structure
- Data parsed into tidy format and prepared for downstream use

Output: Structured dataset with well positions and raw values



Screen 2: Inspect

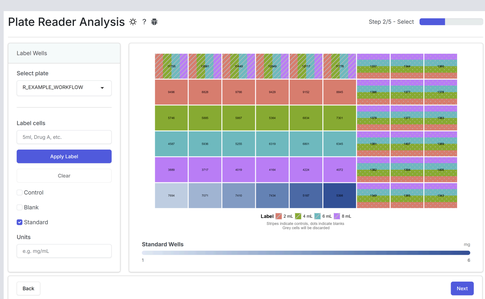
Purpose: Assign wells to experimental groups

Input: Structured dataset

Features & Processing:

- Interactive plate map with color-coded wells
- Assign treatment groups and standard wells
- Link controls and blanks to groups
- Data mapped to plate matrix and grouped via factor levels

Output: Fully labeled dataset with assigned roles



Screen 3: Normalize

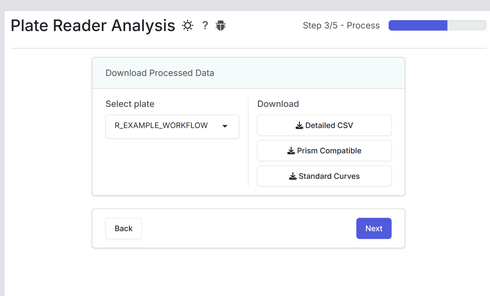
Purpose: Standardize data and enable export

Input: Labeled dataset

Features:

- Calculations:
  - Blank-corrected = value − blank mean
  - Normalized = (corrected / control mean)
- Handles multiple plates and preserves raw and normalized values
- Downloadable outputs: clean CSV, Prism-ready file, and standard wells with predictions

Output: Clean dataset with normalized values



Screen 4: Analysis

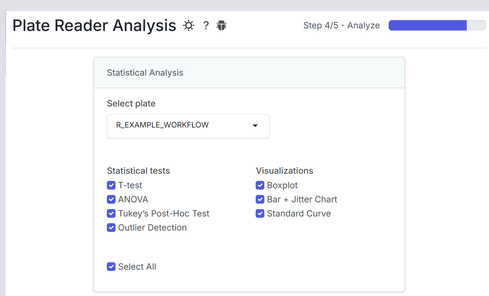
Purpose: Perform exploratory data analysis

Input: Normalized data

Features & Processing:

- User-selected analyses and visualizations
- Significance testing (pairwise t-tests and multigroup ANOVA)
- Outlier detection (IQR method)
- Visualizations: box plots, jittered bar charts, standard curves

Output: Statistical analysis outputs and graphs

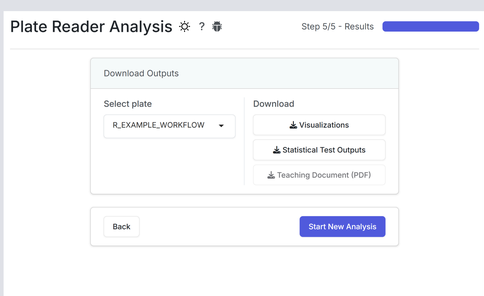


Screen 5: Results

Purpose: Visualize and export results

Input: Statistical results and graphs

Features:

- Download visualizations and analysis results
  - See figures 3-7
- Export teaching document explaining:
  - Normalization steps
  - Statistical methods
  - Interpretation of results

Output: Publication-ready figures, summaries, and a completely transparent report